STAT 516: Homework 7

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(15 points)** Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2}e^{-y}, & -y < x < y,\ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

   Calculate

$$P(X < 1 \mid Y = 3).$$

2. **(15 points)** Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2x, & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

   Calculate

$$P\left(X + Y \leq 1 \mid X \leq \frac{1}{2}\right).$$

3. **(15 points)** Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 24xy, & 0 < x < 1,\ 0 < y < 1 - x, \\ 0, & \text{otherwise.} \end{cases}$$

   Calculate

$$P\left(Y < X \mid X = \frac{1}{3}\right).$$

4. **(13 points)** A fair coin is tossed. If a head occurs, one fair die is rolled. If a tail occurs, two fair dice are rolled. Let Y be the total number of points showing on the die or dice.

   Calculate $\mathrm{E}[Y]$.

5. **(15 points)** Let X and Y denote the values of two stocks at the end of a five-year period. Suppose X is uniformly distributed on the interval $(0, 12)$. Given $X = x$, suppose Y is uniformly distributed on the interval $(0, x)$.

   Calculate

$$\mathrm{Cov}(X, Y).$$

6. **(15 points)** Let X be the loss amount for a policyholder. Suppose X is uniformly distributed on the interval $[0, b]$.

   The policy has a deductible of 180, which means the policyholder is responsible for paying the first 180 of each loss. Therefore, the unreimbursed portion of a loss is

$$\min(X, 180).$$

   If

$$\mathrm{E}[\min(X, 180)] = 144,$$

   calculate b.

7. **(12 points)** Let

$$\mathrm{E}[Y \mid X = x] = 2x, \qquad \mathrm{Var}(Y \mid X = x) = 4x^2,$$

   and suppose X has a uniform distribution on the interval $(0, 1)$.

   Calculate $\mathrm{Var}(Y)$.